MLE-AI-201 · FINAL PROJECT · TEAM 07

Real-Estate Price Prediction and Tier Classification on Bina.az

Decision Tree and Support Vector Machine, implemented from scratch in NumPy.

From-scratch models

Leakage-safe pipeline

Honest scikit-learn benchmark

PROBLEM & DATASET

Two tasks on bina.az listings

Raw sale listings scraped from Azerbaijan’s largest property platform.

TASK A · REGRESSION

Predict the sale price

Trained on log(price) to tame the heavy-tailed price distribution.

TASK B · CLASSIFICATION

Predict the price tier

Binary label: premium vs. standard, derived from the price.

212,750 AZN Tier threshold: the median of the training set only, so no information leaks from validation or test.

DATA HYGIENE

Cleaning, EDA and leakage control

24,917

Duplicates removed

Matched on listing content, plus fixed-bound outlier rules. Mixed string numbers and currency quirks were standardized.

2

Leakage columns dropped

unit_price and total_price arithmetically encode the target, so they never reach a model.

70/15/15

Train / val / test

Deterministic split with seed 42. Scaling and the tier threshold are fit on training data only.

EDA signals

Geographic clustering (lat/lng), room counts, floor area in m², and category mix (Yeni tikili, Torpaq, and others).

MODEL 1 · SRC/DECISION_TREE.PY

Decision tree from scratch

Split criteria

Gini and entropy with information gain; variance (MSE) reduction for regression.

Growth

Recursive build with numeric threshold search, max_depth, min_samples_split and min_samples_leaf.

Correctness

Splits checked against scikit-learn on controlled examples. Test scores match to three decimals.

Validation score by max_depth (gini)



Chosen: gini, depth 20, min leaf 1 (classification). MSE, depth 12, min leaf 5 (regression).

MODEL 2 · SRC/SVM.PY

Pegasos SVM and kernelization

Solver

Regularized hinge-loss primal, solved by Pegasos sub-gradient descent. No QP or SMO solver.

Geometry

Tracks the objective, ‖w‖, margin width 2/‖w‖ and the support vectors where $y(w \cdot x + b) \leq 1$.

Non-linearity

Random Fourier Features approximate an RBF kernel, then the same linear solver runs on them.

Test F1, Task B

0.793	0.850	0.868
Linear Pegasos (scratch)	RFF-SVM (scratch)	sklearn SVC, RBF

BENCHMARK VS SCIKIT-LEARN

Scratch models match the references

Task A · test, log price

Model	RMSE	R^2
Mean baseline	0.781	0.000
Scratch tree	0.340	0.811
sklearn tree	0.340	0.811
Scratch Ridge	0.580	0.448
sklearn Ridge	0.580	0.448

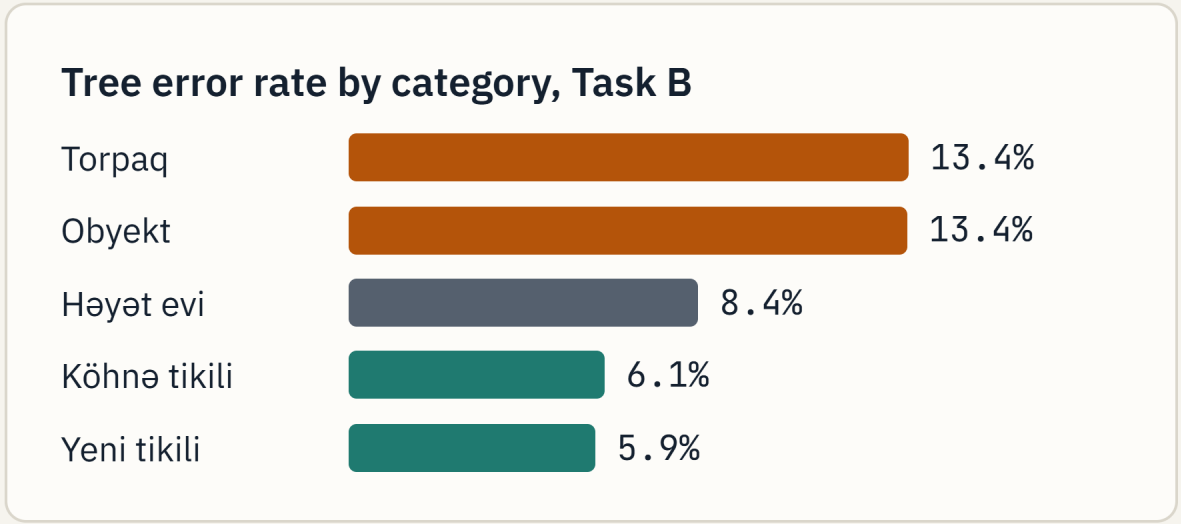
Task B · test

Model	Acc	F1	AUC
Majority class	0.500	0.000	0.500
Scratch tree	0.930	0.930	0.932
sklearn tree	0.931	0.931	0.934
Scratch linear SVM	0.790	0.793	0.868
sklearn SGD	0.791	0.794	0.868
Scratch RFF-SVM	0.849	0.850	0.930
sklearn SVC (RBF)	0.867	0.868	0.949

Tuned on validation only; test used once. 5-fold CV std stays under 0.008 on every model.

DIAGNOSTICS

Where the models fail



Land is hardest

Torpaq regression error is 0.66 in log price, versus 0.14 for new builds.

Extreme listings

Worst misses are land plots with huge areas and inconsistent prices.

Main drivers

Floor area, lat/lng and land size lead the tree importances.

WRAP-UP

Team contributions and next steps

Who did what

Sanan Badalov · data, evaluation, Ridge, report

Oghuz Hasanli · decision tree + tests

Sadigov Sadigov · Pegasos SVM + RFF

Ali Hasanli · sklearn baselines, CV, pipeline

Everyone can defend any part of the code.

Reproducibility

Pinned dependencies and fixed seed 42. One command rebuilds every number and figure from a clean checkout:

```
python -m src.run_all
```

Details in contribution_report.md.

Next steps Geospatial clustering with DBSCAN, and robust ensembles to tighten residual errors on land and luxury listings.